Proposal ID	: **OI2026B-012**
Data Received	: 30/04/2026 11:30

A.M.(<=)	Any	**Moon P.**	Any/Bright
Image Quality		Any seeing os	
Cloudy Cover		Any	

Title of Proposal: ***Mutual Phenomena of Jupiter's satellites***

Telescope **0.60m Boller & Chivens**

Abstract:

Mutual phenomena are events that occur twice during an object's orbital period, taking place when the satellite's orbital plane aligns with the observer's line of sight. In Jupiter's case, this happens every five years and lasts for just over a year. Observational campaigns for the mutual phenomena of the Galilean moons have been conducted since 1974; they remain the only method capable of providing sufficient precision in satellite separation to detect, for instance, tidal evolution. The 2026B marks the beginning of a new series of Jovian mutual phenomena. We have selected nine events occurring across six different nights, focusing primarily on the Galilean satellites.

Category: Solar System: Small Bodies (Asteroids, Comets, Moons, Kuiper Belt, Near Earth Objects)

Time requested: 48 hours.

Minimum time accepted: 32 hours.

Observing Mode: Classical Remote (CR)

Program Type: Regular Proposal (RP)

Principal Investigator (PI): Giuliano Margoti

PI Affiliation: ON

PI Phone/PI Email: (41)988231820 giulianomargoti@gmail.com

Co-authors

Name	Affiliation	E-mail
Felipe Braga Ribas	UTFPR/DAFIS	fribas@utfpr.edu.br
Bruno E. Morgado	OV/UFRJ	bmorgado@ov.ufrj.br
Marcelo Assafin	OV/UFRJ	marcelossafin@gmail.com
Julio Ignacio Bueno de Camargo	ON	camargo@on.br
Altair Ramos Gomes Junior	UFU	altairgomesjr@gmail.com
Rafael Sfair	FEG/UNESP	rafael.sfair@unesp.br
Luan Catani	OV/UFRJ	lcatani@ov.ufrj.br
Viviane Peixoto	OV/UFRJ	vivianepeixoto989@gmail.com
Eros O. Gradovski	ON	erosgradovski@gmail.com

Is there a thesis or dissertation that will benefit from the data?: NO

Optimal dates:

2026-10-25; 2026-10-29; 2026-11-13; 2026-11-15; 2026-12-06; 2026-12-19

Impossible dates:

any other

Instruments:

Imager (CAM 1; 2 or 4)

Detectors	- Ikon 2048 x 2048
Filters	- u Stromgren #1 (50 x 50 mm) - v Stromgren #2 (50 x 50 mm) - b Stromgren #1 (50 x 50 mm) - y Stromgren #1 (50 x 50 mm)

Scientific Justification

Mutual phenomena are events that occur when the orbital plane of an object's satellites aligns with the observer's line of sight. This configuration happens twice per orbital period and can last for over a year, depending on the specific events. In the case of Jupiter, the mutual phenomena of its Galilean satellites will be visible from 2026B to 2028A at the Pico dos Dias Observatory (OPD).

Mutual events represent the most precise method for measuring the relative positions between satellites from ground based observations. The distance between the satellites is not measured directly through astrometry, but rather via photometry. During the phenomenon, when one satellite passes in front of another (occultation, Figure 1) or casts its shadow upon it (eclipse, Figure 2), the total light flux varies over time, producing light curves with distinct morphologies (Figure 3). The depth and shape of the resulting curve depend on the variation in the bodies' relative positions during the event, which are obtained with high precision, smaller them 1 mas, through applied models (Aksnes et al. 1984).

Since 1974, regular campaigns have been conducted to observe these phenomena, primarily for Jupiter's Galileans due to their high brightness (Aksnes et al. 1984). With the accessibility of CCD cameras, this type of observation has become more comprehensive. Several campaigns have already been carried out for the satellites of Jupiter (Aksnes et al. 1984; Vasundhara 1994; Arlot 1996, 2002; Emelyanov & Gilbert 2006; Emelyanov 2009; Dias-Oliveira et al. 2013; Arlot et al. 2014; Catani et al. 2023), Saturn (Aksnes et al. 1984; Arlot et al. 2012), and Uranus (Assafin et al. 2009; Arlot et al. 2013). Since the foundation of the OPD, Brazil has actively contributed to this international collaboration. Data obtained within national territory have resulted in four Master's dissertations (Braga-Ribas 2009; Dias-Oliveira 2011; Morgado 2015; Catani 2023) and two Doctoral theses (Morgado 2019; Santos-Filho 2019), in addition to publications based exclusively on OPD data (Dias-Oliveira et al. 2013; Assafin et al. 2009) or in partnership with the collaboration (Arlot et al. 2014; Catani et al. 2023; Arlot et al. 2012; Arlot et al. 2013).

Due to the high precision achieved, these observations are the only ground-based measurements that allow for the determination of secular effects of certain perturbations in planetary satellite systems, such as those caused by tidal effects (Lainey et al. 2009). These perturbations, combined with orbital resonances, played a crucial role in the current configuration of the giant planets' primary satellites (Batygin & Morbidelli 2020). The orbital plane of the Galilean satellites is not yet perfectly aligned with Earth; therefore, during this period, we will observe almost exclusively eclipses, with only three selected occultations. The phenomena are classified as total (T), partial (P), annular or penumbral (A), and grazing (R), for both occultations (OC) and eclipses (EC). Since the Galilean satellites are highly luminous, Europa (502) mag: 6.1, Io (501) mag: 5.9, Ganymede (503) mag: 5.5, and Callisto (504) mag: 6.7, we are submitting this request to the B&C, IAG. A total of 9 events are to be observed over 6 nights; diagrams of the first occultation can be seen in Figure 4.

The use of a methane filter (wavelength 889 nm, narrow-band 18 nm) blocks light reflected by Jupiter's atmosphere while allowing light reflected by the satellites to pass will be necessary. This reduces scattered light on the CCD without compromising the satellites' brightness. Combined with our digital coronagraphy technique, this allows for high-quality results using IAG data, as demonstrated in previous studies (Arlot et al. 2014; Catani et al. 2023; Arlot et al. 2012; Assafin et al. 2009). Through our participation in previous campaigns for Uranus, Saturn, and Jupiter, we have developed various observation and image processing techniques, including image differencing. This allows us to create a template of Jupiter's light and subtract it from the science images, thereby improving photometric quality and the resulting models for the event (Catani 2023).

Experimental Design

The primary objective of this proposal is to observe the mutual phenomena of the Galilean satellites (Io, Europa, Ganymede, and Callisto) occurring in the second half of 2026. By obtaining photometric light curves of the flux drops, we aim to determine the precise relative positions between the satellites to refine existing orbital models and better understand the interaction between orbital dynamics and observed geological behaviors, with a specific emphasis on tidal effects.

These targets were selected due to their high brightness, as fainter satellites would require larger apertures, such as the 1.6m Perkin-Elmer (P&E) telescope at OPD. However, as evidenced by projects P-008 and P-009 (2025A/B), observations of such events are significantly hindered without a methane filter to mitigate Jupiter's scattered light. Consequently, our targets have been optimized for the B&C IAG telescope.

Several filters were applied to select our mutual phenomena targets. From a total of over 27,000 potential events occurring at the OPD, we first isolated those involving only the Galilean moons. We further constrained the selection to events occurring between 19:00 and 05:00 local time, with an elevation higher than 25 degrees, a flux drop exceeding 10%, and a duration of more than 2 minutes.

Our strategy employs a methane filter to suppress atmospheric reflection from the planet. We will maintain at least one calibration star in the field and utilize the shortest possible exposure times for the IkonL. As these events are time-critical and occur on specific nights, there are no restrictions regarding cloud cover or moon phase. Observations will proceed regardless of weather conditions, provided the night technician grants safety clearance for telescope operation.

As a backup project, we will perform direct astrometry of the Galilean satellites, including close encounters prior to occultations or eclipses. Classical astrometry for these objects is non trivial due to the extreme brightness of Jupiter, with standard methods yielding precision around 100 mas (Kiseleva et al. 2008; Narizhnaya 2016). To overcome this, we will apply an alternative technique called Consecutive Image Combination. Since individual frames rarely contain enough reference stars, we will use bright stars ($mag \leq 12$) to align series of consecutive images, improving the Signal-to-Noise Ratio (SNR) to reveal a sufficient number of background reference stars. Initial tests indicate this technique can achieve a precision of approximately 20 mas (Morgado 2019). This backup plan also extends to observing other Solar System objects for astrometric or rotational studies, all of which are feasible using the IkonL camera and provide full-night coverage.

Regarding precision, our astrometric results for mutual phenomena will depend on the event depth, which ranges from 13% to 53% in our targets. We require a photometric noise level of at least 10% of the flux drop (0.01 to 0.05), resulting in a minimum SNR of 50. For direct Galilean astrometry, we aim for an SNR of 800, while for other targets ($mag < 18$), we expect to reach an SNR greater than 200.

Technical Description

For our observations, we request the use of the B&C IAG telescope equipped with an IkonL camera and a filter set including Methane (wavelength 889 nm, narrow-band 18 nm), Bessel (V, R, I), and a Clear filter. This specific setup is essential to accommodate very short exposure times, under one second in some instances, providing the high temporal cadence required to achieve superior precision in our mutual phenomena models. For backup astrometry and rotational studies, we may employ alternative filters to perform color analysis or to enhance the contrast between the targets and the stellar background. Given that these events are time-critical and occur on fixed dates, our observations are independent of sky quality or moon phase.

Regarding our signal-to-noise ratio (SNR) goals, we intend to obtain an SNR of at least 50 for the mutual phenomena events, which may necessitate exposure times shorter than 1 second. For the direct astrometry of the Galilean satellites, we plan to use exposure times between 4 and 20 seconds to achieve an SNR greater than 800. For our secondary rotation and astrometry targets, we expect to use exposure times ranging from 90 to 200 seconds to reach an SNR above 200.

Our mutual phenomena targets selection were designed to prioritize events that yield the most robust results, culminating in 9 specific events distributed across 6 nights: October 26, October 30, November 14, November 16, December 7, and December 20, 2026.

Finally, the data analysis will be conducted using proprietary reduction pipelines. These include the PRAIA/photometr package (Assafin 2023) for photometric reduction, and the SORA Python package (Gomes-Júnior et al. 2022) for light curve reduction and the determination of event instants.

Figures and Tables

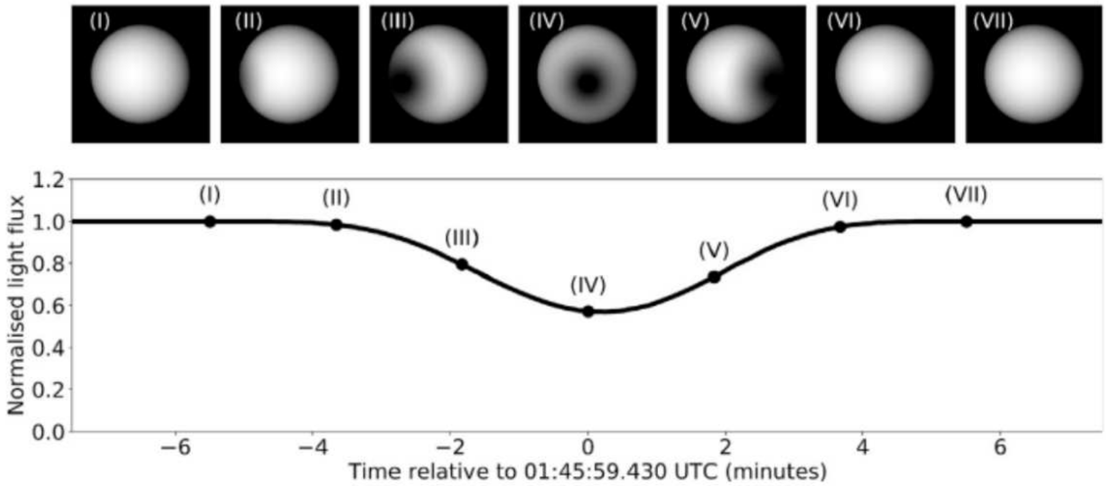

Figure 1: Simulation of an eclipse. The upper panel shows the eclipsed surface brightness of a satellite, while the lower panel displays the resulting light curve from total light fluxes. Images taken before and after the event provide the instrumental albedo of the satellites used in the simulations. Satellite sizes are integrated into the modeling, allowing for the validation of the assumed sizes of the inner satellites. Satellite ephemerides provide the initial relative positions as starting points for the simulations. The resulting light curves are normalized to 1 outside the event.

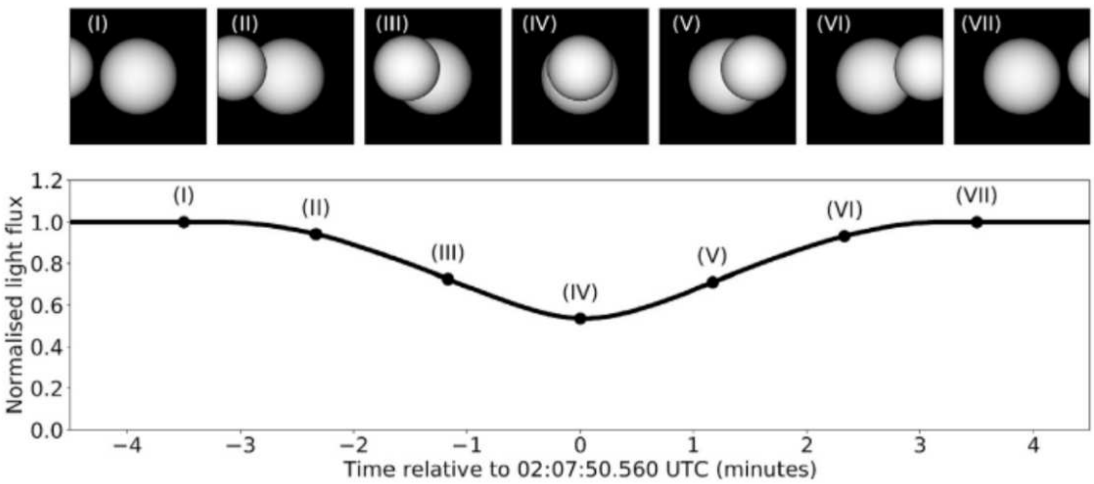

Figure 2: Simulation of the surface brightness of satellites during an occultation (upper panel) and the resulting normalized light curve from total light fluxes (lower panel). Images taken before and after the event provide the instrumental albedo of the satellites involved in the simulations. Satellite sizes are integrated into the modeling, while satellite ephemerides provide the initial relative positions as starting points for the simulations. The generated curves are normalized to 1 outside the event.

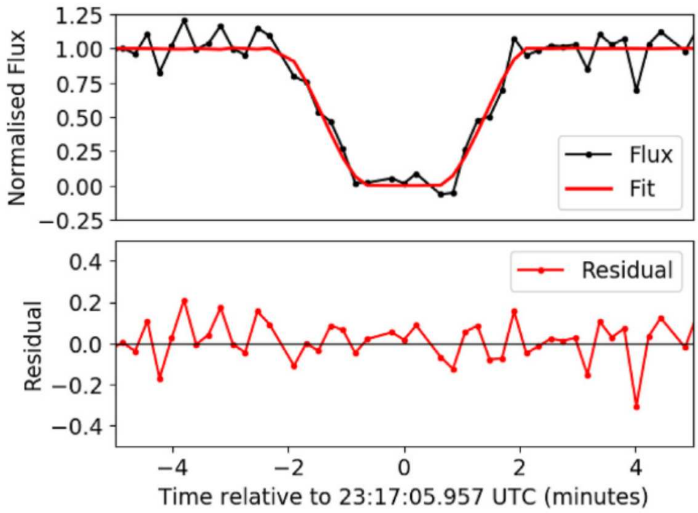

Figure 3: Upper panel: fitted light curve (red line) over the observed light curve (black points) of the eclipse of Amalthea (V = 14.5) by Ganymede on March 2nd, 2015, observed at the P&E (OPD) (Morgado et al. 2019). The residuals are shown below. The light curves generated from simulated fluxes are fitted to the observed curves through an iterative chi-square minimization process until convergence is reached. From the final fitted curve—associated with the resulting relative motion (relative positions and velocities)—we derive the central instant (time of closest approach) and the impact parameter (relative position at the central instant), expressed as ephemeris offsets in right ascension and declination. This curve, published in Morgado et al. (2019), was obtained using the IkonL camera and a methane filter, with an exposure time of 4s and a dead time of 4s (8s cycle), resulting in a precision of 48 mas in relative positions with an SNR = 20.

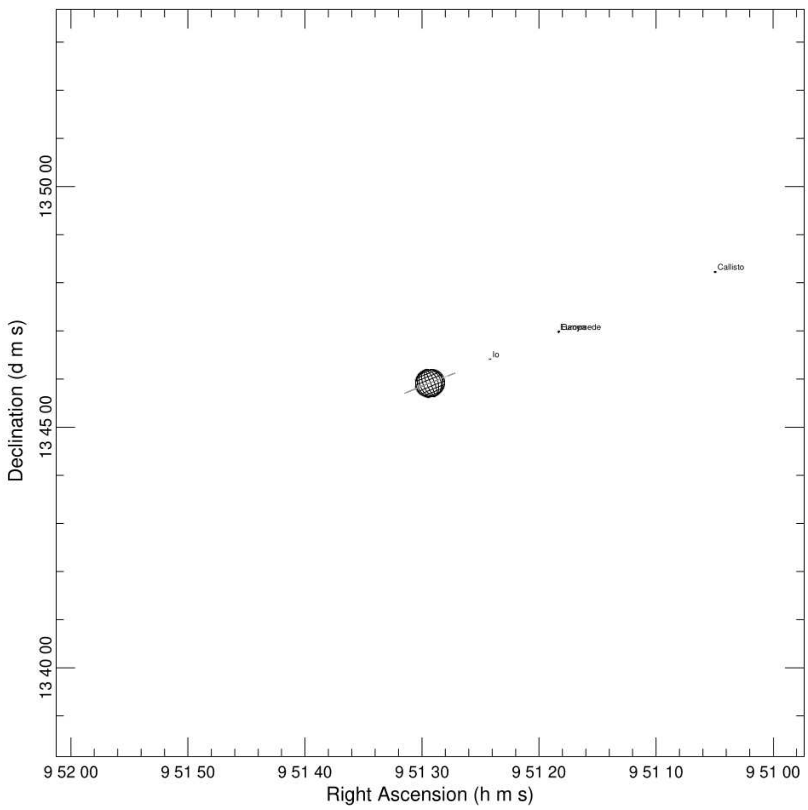

Figure 4: Event diagram for 2026-11-14 07:51:43.960, showing a partial occultation between Europa and Ganymede.

References

- Aksnes K., Franklin F., Millis, R. et al. 1984, Astronomical Journal, 89, 2
- Arlot, J.-E. 1996, Astronomy and Astrophysics, 314, 312
- Arlot, J.-E. 2002, Astronomy and Astrophysics, 383, 719
- Arlot, J.-E., Emelyanov, N. V., Lainey, V. 2012, Astronomy and Astrophysics, 544, A29
- Arlot, J.-E., Emelyanov, N. V., Aslan, Z. 2013, Astronomy and Astrophysics, 557, A4
- Arlot, J.-E., Emelyanov, N., Varfolomeev, M. I. 2014, Astronomy and Astrophysics, 572, A120
- Assafin, M., Vieira-Martins, R., Braga-Ribas, F. 2009, The Astronomical Journal, 137, 4046
- Assafin, M. 2023, Planetary and Space Science, 239, 105816
- Batygin, K., Morbidelli, A. 2020, AphJ, 894, 143
- Braga Ribas, F. 2009, ”Fenômenos Mútuos entre os Satélites de Urano”, Dissertação de Mestrado, Observatório do Valongo, Fevereiro/2009
- Camargo,J.I.B., Magalhães,F.P.,Vieira-Martins,R. et al. 2015, Astronomy and Astrophysics,582, A8
- Catani, L. M. 2023, ”Análise dos Fenômenos Mútuos de 2021 dos Satélites Galileanos e Internos de Júpiter” Dissertação de Mestrado, Observatório do Valongo, Março/2023
- Catani, L. M., Assafin, M., Morgado, B. E. 2023, Monthly Notices of the Royal Astronomical Society, 526, 6145
- Dias-Oliveira, A. 2011, ”Fenômenos Mútuos entre os Satélites Galileanos de Júpiter”, Dissertação de Mestrado, Observatório do Valongo, Fevereiro/2011
- Dias-Oliveira, A., Vieira-Martins, R., Assafin, M. 2013, Monthly Notices of the Royal Astronomical Society, 432, 225
- Emelyanov, N. V., Gilbert, R. 2006, Astronomy and Astrophysics, 453, 1141
- Emelyanov, N. V. 2009, Monthly Notices of the Royal Astronomical Society, 394, 1037
- Gomes-Júnior, A. R., Morgado, B. E., Benedetti-Rossi, G. 2022, Monthly Notices of the Royal Astronomical Society, 511, 1167
- KISELEVA, T. P. et al. Planetary and Space Science, v. 56, n. 14, p. 1908 1912, 2008.
- Lainey, V., Arlot, J.-Eudes et al. 2009, Nature, 459, 957
- Morgado, B. 2015, ”Astrometria dos Satélites Galileanos de Júpiter”, Dissertação de Mestrado, Observatório do Valongo, Julho/2015
- Morgado, B. 2019, ”Estudo Astrométrico dos Satélites Galileanos de Júpiter”, Tese de Doutorado, ObservatórioNacional, Setembro/2019
- Morgado, B., Vieira-Martins, R., Assafin, M. et al. 2019, Planetary and Space Science, 179, 104736
- NARIZHNAYA, N. V. Solar System Research, v. 50, p. 344 351, 2016.
- Santos Filho, S. 2019, ”Métodos Alternativos na Astrometria dos Satélites Principais de Urano”, Tese de Doutorado, Observatório do Valongo, Outubro/2019
- Vasundhara, R. 1994, Astronomy and Astrophysics, 281, 565

Previous missions of this proposal to this telescope

This proposal has not been previously submitted.

Previous missions of other proposals to this telescope

No previous proposals have been submitted to this telescope by the PI

Previous results in the field by the Principal Investigator

The PI does not yet have any published work specifically regarding mutual phenomena.

PI and Co-Is publications relevant to this proposal

The co-authors of this proposal have extensive experience with mutual phenomena, having published the following Master's dissertations and PhD theses:

- Braga Ribas, F. 2009, ”Fenômenos Mútuos entre os Satélites de Urano”, Dissertação de Mestrado, Observatório do Valongo, Fevereiro/2009
- Catani, L. M. 2023, ”Análise dos Fenômenos Mútuos de 2021 dos Satélites Galileanos e Internos de Júpiter” Dissertação de Mestrado, Observatório do Valongo, Março/2023
- Dias-Oliveira, A. 2011, ”Fenômenos Mútuos entre os Satélites Galileanos de Júpiter”, Dissertação de Mestrado, Observatório do Valongo, Fevereiro/2011
- Morgado, B. 2015, ”Astrometria dos Satélites Galileanos de Júpiter”, Dissertação de Mestrado, Observatório do Valongo, Julho/2015
- Morgado, B. 2019, ”Estudo Astrométrico dos Satélites Galileanos de Júpiter”, Tese de Doutorado, ObservatórioNacional, Setembro/2019
- Santos Filho, S. 2019, ”Métodos Alternativos na Astrometria dos Satélites Principais de Urano”, Tese de Doutorado, Observatório do Valongo, Outubro/2019

Furthermore, the following works related to mutual phenomena, involving collaboration with the co-authors, have been published in international journals; the most relevant ones are:

- Arlot, J.-E., Emelyanov, N. V., Lainey, V. 2012, Astronomy and Astrophysics, 544, A29
- Arlot, J.-E., Emelyanov, N. V., Aslan, Z. 2013, Astronomy and Astrophysics, 557, A4
- Arlot, J.-E., Emelyanov, N., Varfolomeev, M. I. 2014, Astronomy and Astrophysics, 572, A120
- Assafin, M., Vieira-Martins, R., Braga-Ribas, F. 2009, The Astronomical Journal, 137, 4046
- Catani, L. M., Assafin, M., Morgado, B. E. 2023, Monthly Notices of the Royal Astronomical Society, 526, 6145
- Dias-Oliveira, A., Vieira-Martins, R., Assafin, M. 2013, Monthly Notices of the Royal Astronomical Society, 432, 225
- Emelyanov, N. V., Gilbert, R. 2006, Astronomy and Astrophysics, 453, 1141
- Morgado, B., Vieira-Martins, R., Assafin, M. et al. 2019, Planetary and Space Science, 179, 104736

Objects List.

Object	A.R.(h,m)	Dec.(deg)	Mag.	Band	Type	Comments
Io (501)	09:43:13.79	+14:24:05.40	5.9	V	Sci	Europa eclipses Io at 2026-10-26 08:08 UTC
Callisto (504)	09:45:14.76	+14:14:41.50	6.6	V	Sci	Europa eclipses Callisto at 2026-10-30 07:31 UTC
Ganymede (503)	09:51:18.33	+13:47:00.00	5.3	V	Sci	Europa occult Ganimede at 2026-11-14 07:51 UTC
Callisto (504)	09:52:03.54	+13:43:25.30	6.6	V	Sci	Amalthea eclipses Callisto at 2026-11-16 06:53 UTC
Callisto (504)	09:52:04.66	+13:43:19.70	6.6	V	Sci	Io eclipses Callisto at 2026-11-16 07:46 UTC
Callisto (504)	09:52:04.89	+13:43:18.50	6.6	V	Sci	Adrastea eclipses Callisto at 2026-11-16 07:57 UTC
Ganymede (503)	09:56:35.17	+13:24:46.30	5.2	V	Sci	Europa eclipses Ganymede at 2026-12-07 04:56 UTC
Ganymede (503)	09:56:11.33	+13:30:33.20	5.2	V	Sci	Europa eclipses Ganymede at 2026-12-20 03:46 UTC
Io (501)	09:56:29.66	+13:28:47.00	5.5	V	Sci	Callisto eclipses Io at 2026-12-20 07:27 UTC